Jarod Kraven R. Manikan

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Education

National University — Calamba, Laguna
Bachelor of Science in Computer Engineering — Aug 2022 – July 2026
Relevant Coursework: Cybersecurity, Embedded Systems, Microprocessors

STI College — Santa Rosa, Laguna
Senior High School – IT in Mobile App and Web Development — Sep 2020 – Aug 2022
Graduated with Honors

Skills

- **Programming Languages:** C++, Python, Java, JavaScript, HTML/CSS, MATLAB, Verilog
- **Software:** AutoCAD, VS Code, Proteus, Multisim
- **Hardware:** Arduino, Raspberry Pi, PCB Design, Logic Design, Microcontroller, Oscilloscope, PLC
- **Other Skills:** Networking Fundamentals, Troubleshooting, Computer Servicing, ServiceNow

Experience

IT Service Delivery Intern, Ivoclar Vivadent, Inc. – Cabuyao, Laguna — Mar 2026 – May 2026
- Reduced recurring IT downtime for on-site staff by performing proactive computer hardware maintenance and network troubleshooting across daily operations.
- Improved IT service request turnaround by organizing, updating, and monitoring tickets in ServiceNow, streamlining overall ticket management.
- Strengthened IT support reliability by assisting in quality assurance checks and identifying workflow gaps for system improvements.
- Developed a Java-based QR/barcode canteen tracking system that automated employee meal access monitoring and improved accuracy during the company's facility transition.

Projects

Development of IoT – Based Rainwater Purification System for Aeta Community — Mar 2025 – April 2026
- Built a solar-powered IoT water purification system using Raspberry Pi 5, RO, UV, and relay-controlled pumps to automate sequential filtration.
- Integrated pH, turbidity, and TDS sensors via ADC for real-time water quality monitoring and ML-based potability classification.
- Developed a local web dashboard for live visualization and control; validated output with a DENR-accredited lab to meet Philippine drinking water standards.
- Co-authored the research paper, presented the project findings at the 18th International Conference on Electronics, Computer and Computation (ICECCO 2026), and published the work in IEEE Xplore.

Peltier Based Atmospheric Water Generator for Sustainable Drinking Water Supply — Nov 2024 – Mar 2025
- Designed small-scale atmospheric water generator using Peltier modules and copper coil condensation.
- Automated multi-stage condensation, filtration, UV sterilization using Arduino UNO and ESP8266.
- Integrated temperature, humidity, pH, turbidity, and TDS sensors for water quality monitoring.
- Implemented relay-based pump control and storage management for automated operation (~800 L/hr system)

AC/DC Multi-Output Power Supply
- Designed and fabricated a custom PCB power supply converting 220V AC to regulated 3V, 5V, and 12V DC using bridge rectifier, filtering stages, and voltage regulators.
- Implemented protection and heat management for stable multi-load operation across embedded and sensor modules.
- Validated output stability with multimeter/oscilloscope testing to ensure reliable power delivery for IoT and microcontroller systems.

Certifications

Computer System Servicing NC II – TESDA — Oct 2025
Generative AI Workshop – Cognizant — Oct 2025
TOEIC Listening & Reading: 900/990 — Dec 2025
Safety Officer 2 Certifications – BOSH — Jan 2026